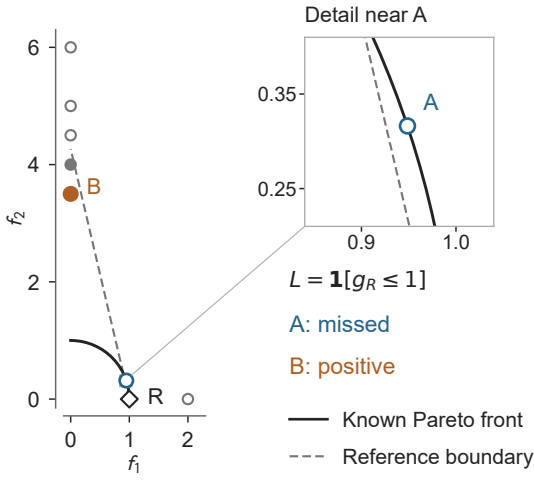
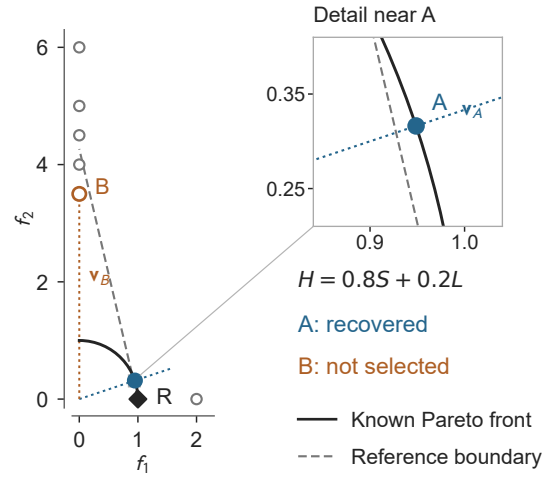


(a) Reference-solution PBI

(b) Hybrid PBI ($t = 0.20$)

Solution	Reference label L	Score S	Score H	Hybrid group
A	0	0.500	0.400	Positive
B	1	0.222	0.378	Non-positive

Filled: positive group Open: non-positive group Diamond: reference R

Constructed example

Figure 1: Illustration of directional information correcting representative-based grouping. The same eight constructed objective vectors are shown in both panels for minimization over $f_1, f_2 \geq 0$ and $f_1^2 + f_2^2 \geq 1$. The unit quarter-circle is the known Pareto front and is used only to assess the illustrated decisions; it is not supplied to either scoring rule. (a) Given the reference $R = (1, 0)$ and $\delta = 0.234375$, the representative-based classifier assigns $L_A = 0$ and $L_B = 1$, although $A = (3/\sqrt{10}, 1/\sqrt{10})$ lies on the front and $B = (0, 3.5)$ lies outside it. The dashed curve is the computed reference classification boundary. (b) With $\theta = 5$ and budget progress $t = 0.20$, directional scores enter $H = 0.8S + 0.2L$. The top two of eight scores select A and R , excluding B . Dotted lines show the directions associated with A and B ; matched insets enlarge the neighborhood of A . Filled and open markers denote membership and non-membership in the respective positive groups; the diamond denotes the reference. The bottom table reports the focal scores. All labels follow the current numerical implementation, including its treatment of points on the classification boundary. This is a classification example with a prescribed reference and parameters, not an experimental result or a validation of reference selection. It illustrates an early-stage correction; cross-label reversal is not guaranteed throughout the budget.